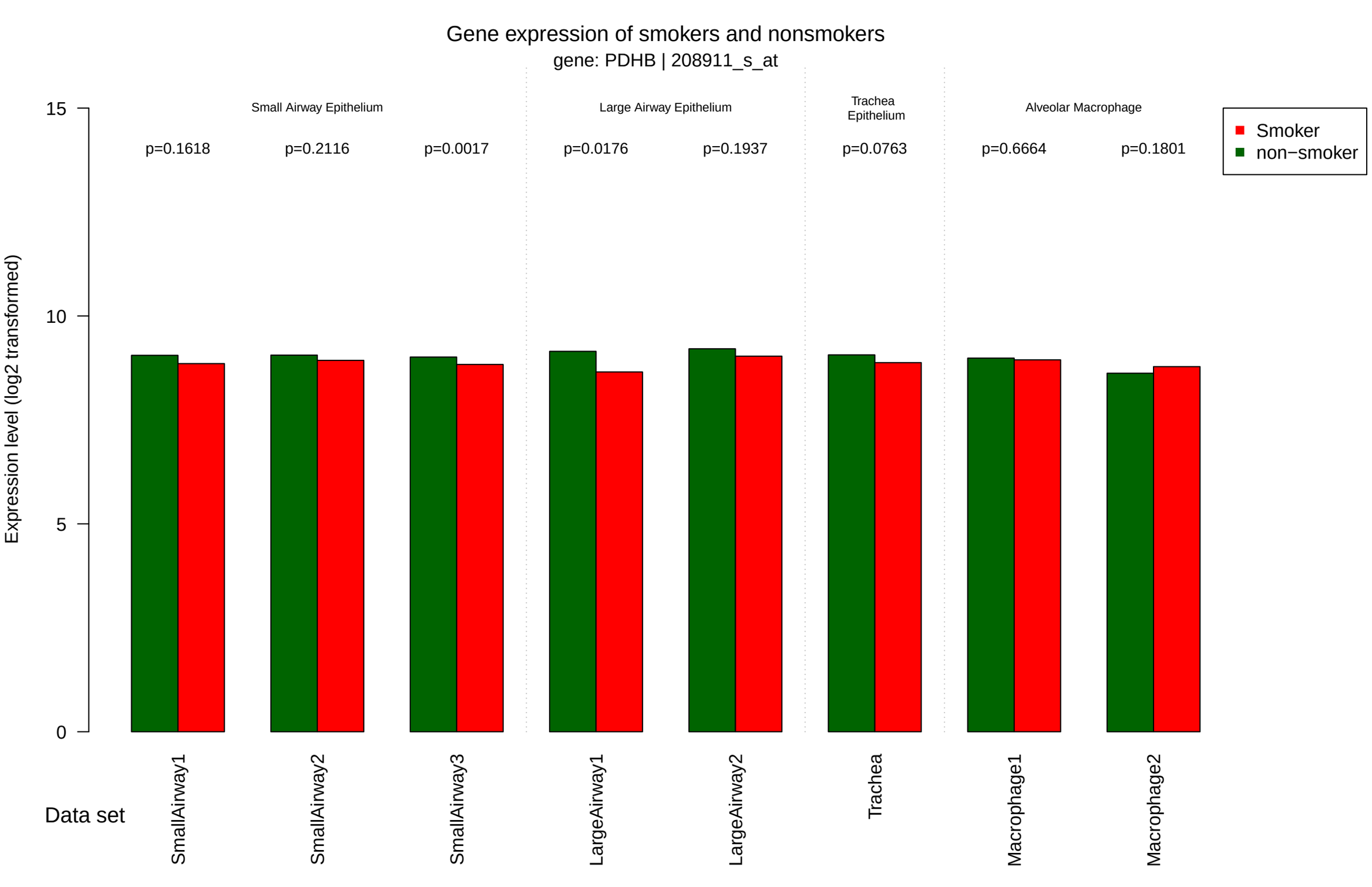
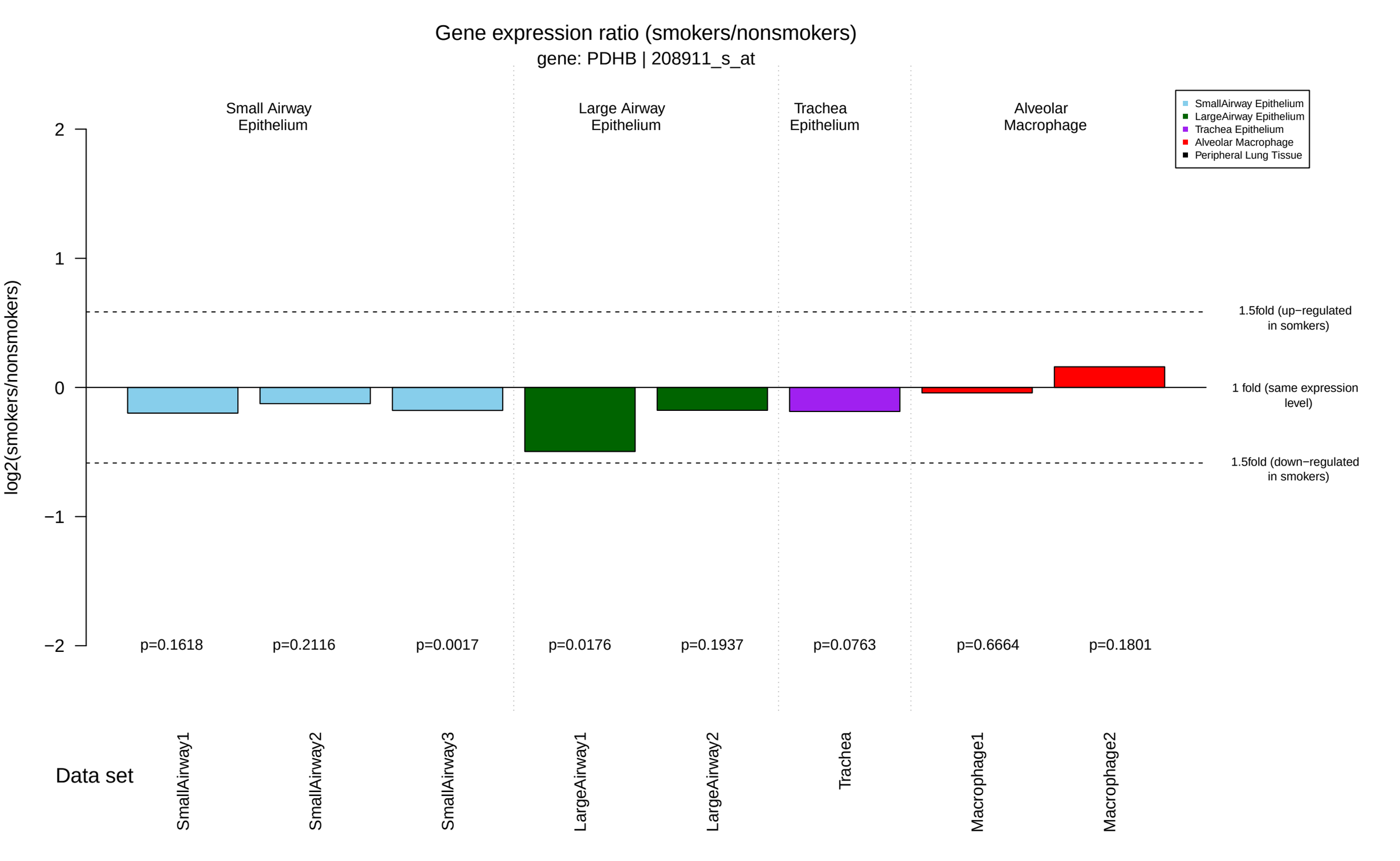
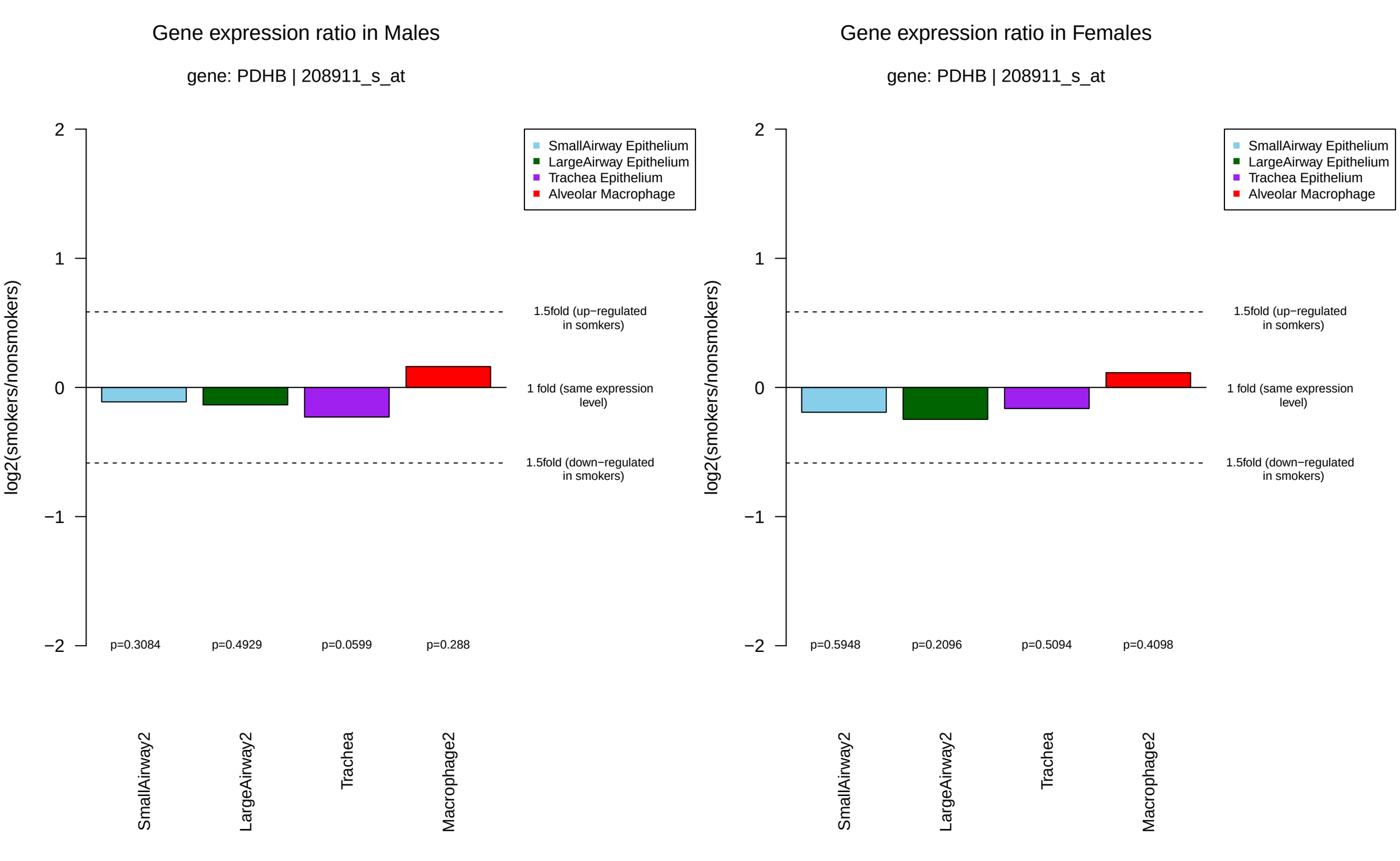
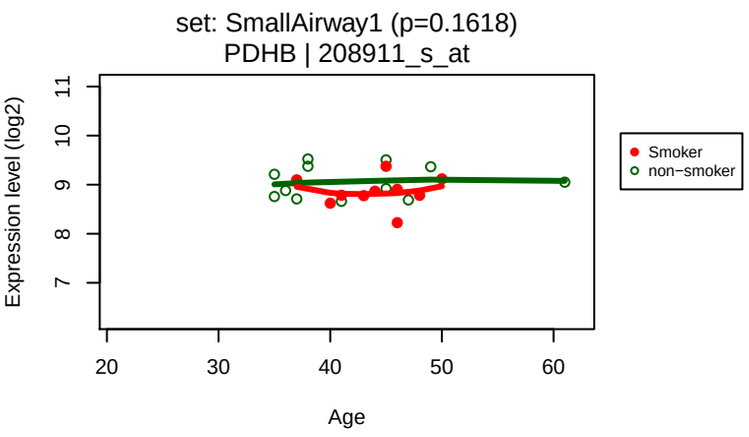
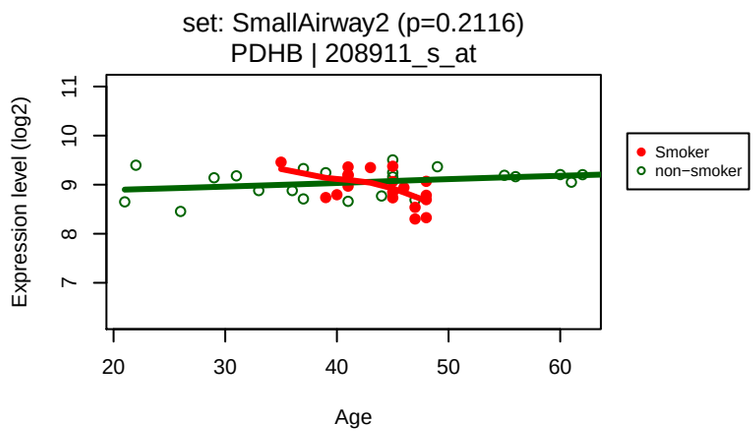
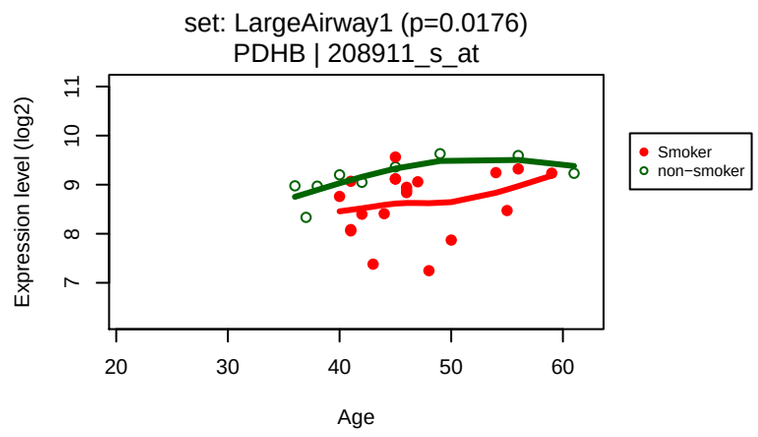

Gene expression of smokers and nonsmokers
gene: PDHB | 208911_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1618
p=0.2116
p=0.0017
p=0.0176
p=0.1937
p=0.0763
p=0.6664
p=0.1801
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

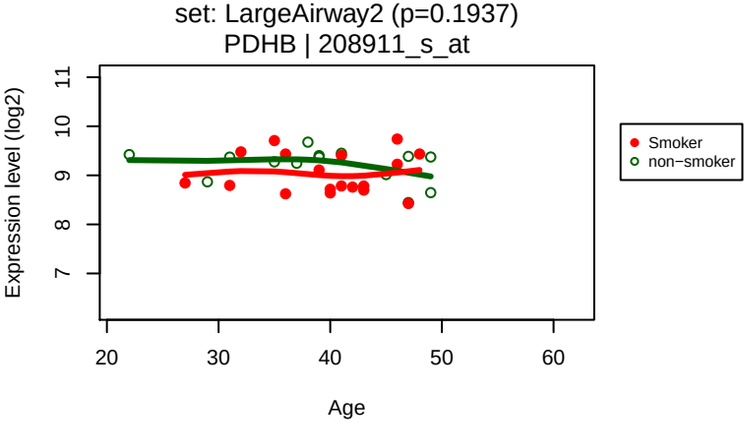

Gene expression ratio (smokers/nonsmokers)
gene: PDHB | 208911_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1618
p=0.2116
p=0.0017
p=0.0176
p=0.1937
p=0.0763
p=0.6664
p=0.1801
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

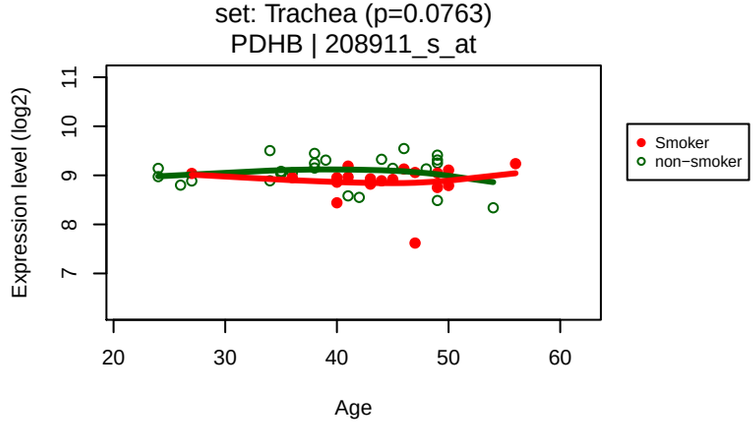

Gene expression ratio in Males
gene: PDHB | 208911_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3084
p=0.4929
p=0.0599
p=0.288
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDHB | 208911_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5948
p=0.2096
p=0.5094
p=0.4098
SmallAirway2
LargeAirway2
Trachea
Macrophage2

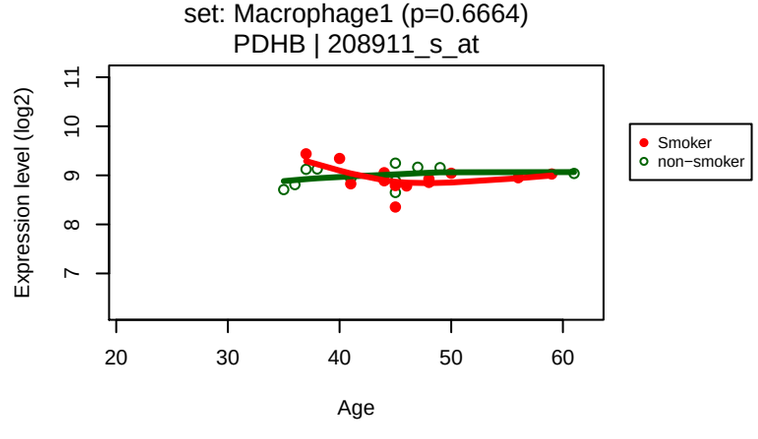

set: SmallAirway1 (p=0.1618)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker

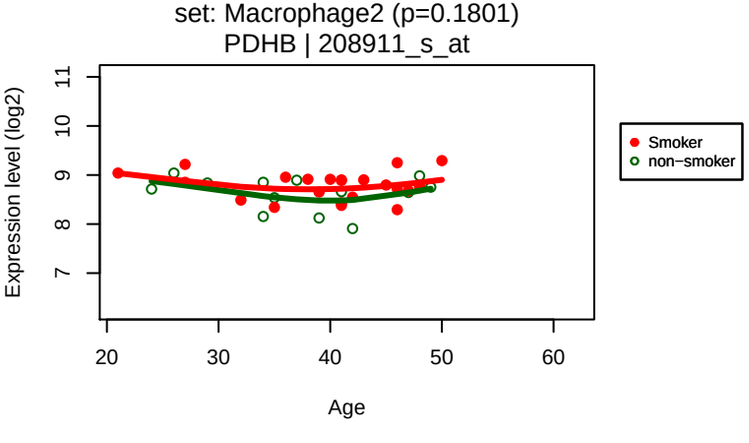

set: SmallAirway2 (p=0.2116)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker

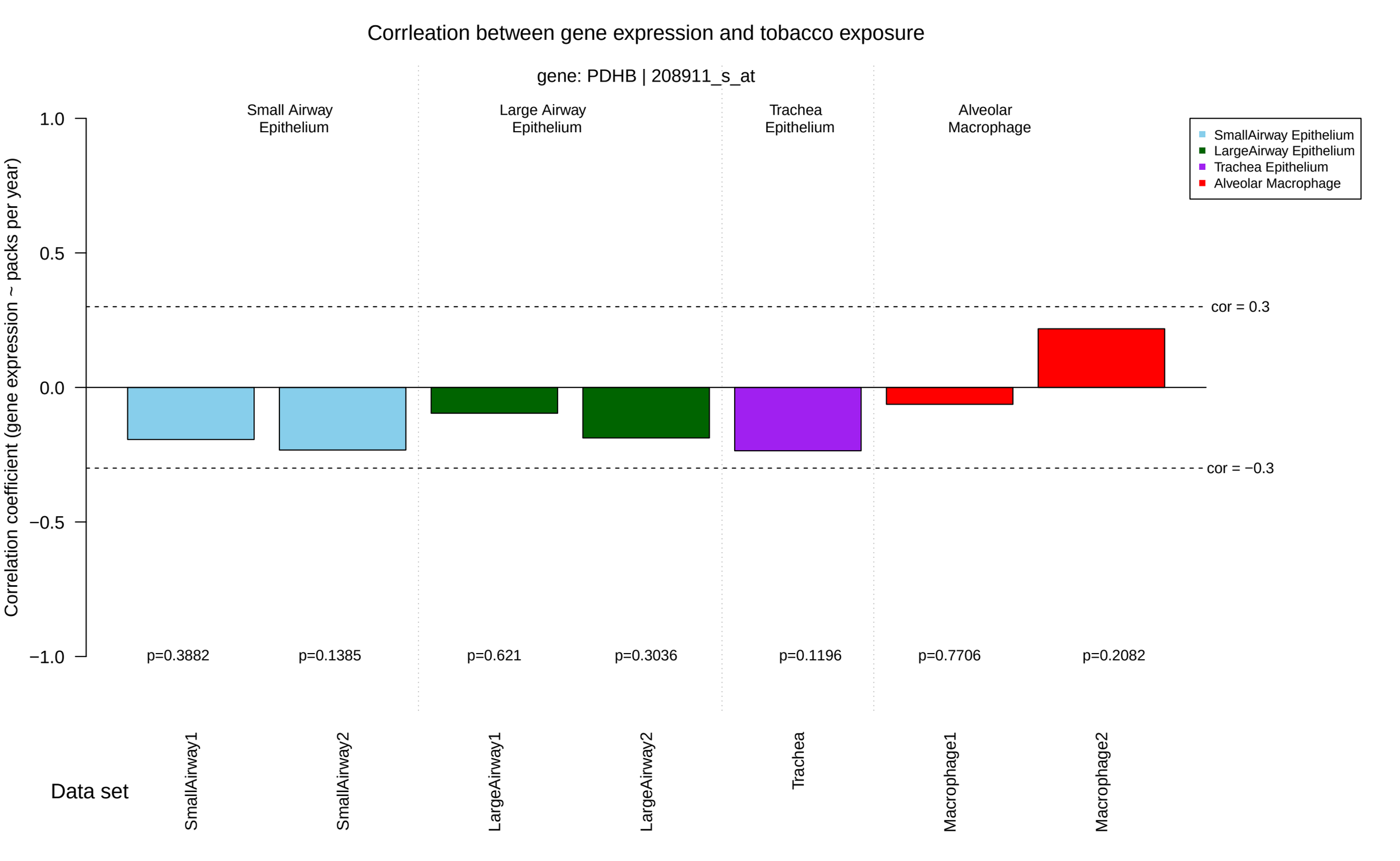

set: LargeAirway1 (p=0.0176)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1937)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0763)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6664)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1801)
PDHB | 208911_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDHB | 208911_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3882
p=0.1385
p=0.621
p=0.3036
p=0.1196
p=0.7706
p=0.2082
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2